
TVA Flowmeter for Saturated and Superheated Steam Service

Description

The Spirax Sarco TVA flowmeter is designed for use on saturated and superheated steam (with the dedicated pressure sensor kit) and operates on the target principle, by measuring the force produced on a moving cone by the fluid flow. This force is then converted into density compensated mass flowrate and is transmitted via a single loop powered 4–20 mA and pulsed output. TVA flowmeters also incorporate a totalised flow function and EIA 232C (RS 232) or EIA 485C (RS485) Modbus communications.

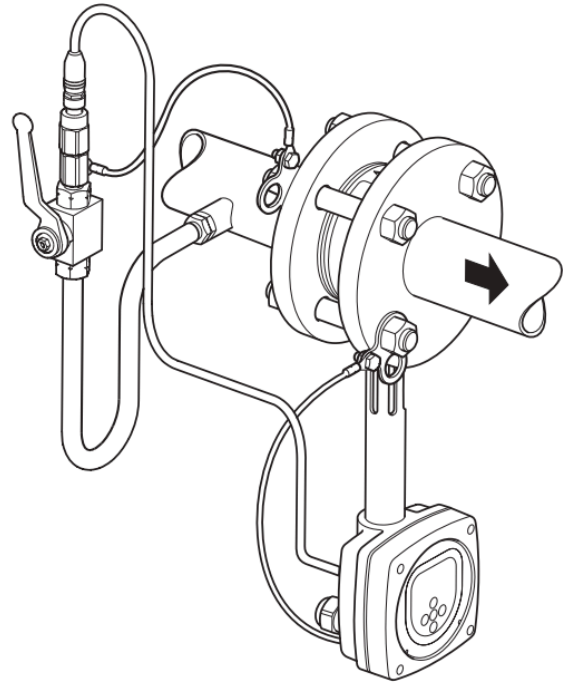
Sizes and pipe connections

DN50, DN80 and DN100.

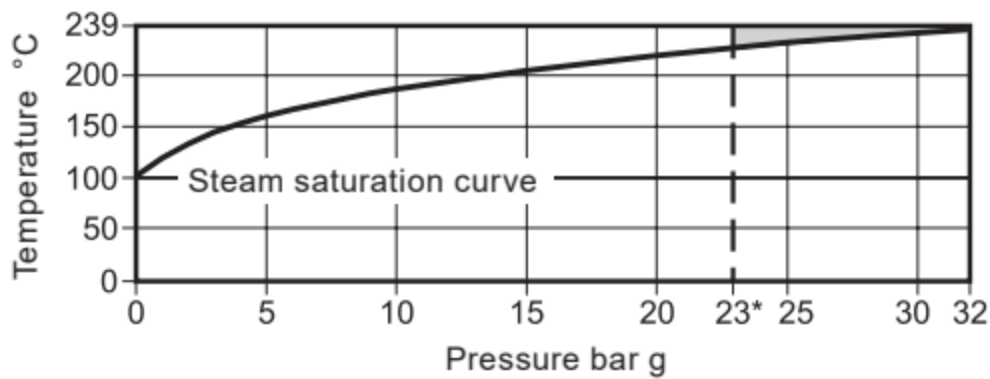
The TVA flowmeter is of wafer design, suitable for fitting between the following flanges:

- EN 1092 PN16, PN25 and PN40
- BS 10 Table H
- ASME B 16.5 Class 150 and Class 300
- Japanese Industrial Standard JIS 20
- Korean Standard KS 20

Note: Spirax Sarco TVA flowmeters should be installed in pipework manufactured to BS 1600, ASME B 36.10 Schedule 40 or EN 10216-2 / EN 10216-5 equivalent. For systems with different standards or schedules, please contact Spirax Sarco.



Pressure/temperature limits



The product should not be used in the indicated region due to software limitations.

Maximum design pressure	32 bar g at 239°C
Maximum design temperature	239°C
Minimum design temperature	0°C (non-freezing)
Maximum operating pressure — horizontal, superheated steam	23 bar g at 239°C
Maximum operating pressure — horizontal, saturated steam	32 bar g at 239°C
Maximum operating pressure — vertical, saturated steam only	7 bar g at 170°C
Minimum operating pressure	0.6 bar g
Maximum operating temperature (saturation)	239°C
Minimum operating temperature	0°C (non-freezing)
Maximum electronics ambient temperature	55°C
Maximum electronics humidity level	90% RH (non-condensing)
Maximum cold hydraulic test pressure	52 bar g

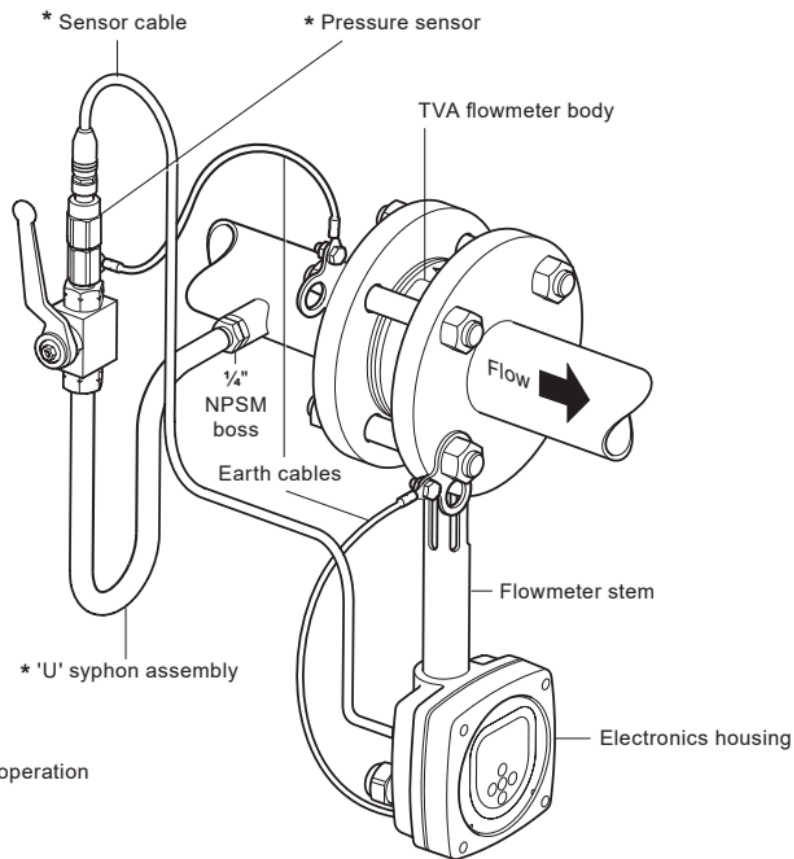
High pressure syphon tube assembly

Maximum design pressure	80 bar g
Maximum design temperature	450°C
Maximum working conditions	60 bar g at 450°C

Pressure sensing kit

Maximum operating temperature	125°C
Minimum operating temperature	0°C (non-freezing)
Maximum operating pressure	50 bar g
Maximum ambient temperature (cable and connector)	70°C

Materials



* Items required for superheat operation.

Unit	Part	Material
TVA	Flowmeter body	Stainless steel S.316, 1.4408 CF8M
	Internals	431 S29 / S303 / S304 / S316
	Spring	Inconel X750 or equivalent
	Flowmeter stem	Stainless steel 300 series
	Electronics housing	Aluminium LM25
Pressure sensing kit	Cable	Polyvinyl chloride (PVC)
	Sensor housing	AISI 304 stainless steel 1.4301
	Sensor	AISI 630 stainless steel 1.4542
	O-ring	Nitrile Butadiene Rubber (NBR)
	Adaptor	AISI 431 stainless steel 1.4057
High pressure syphon tube assembly	Tube	Carbon steel BS 3602: Part 1 1987 CFS 360 (zinc plated/passivated)
	Valve body	Carbon steel
	Valve seat	PEEK / Polymain

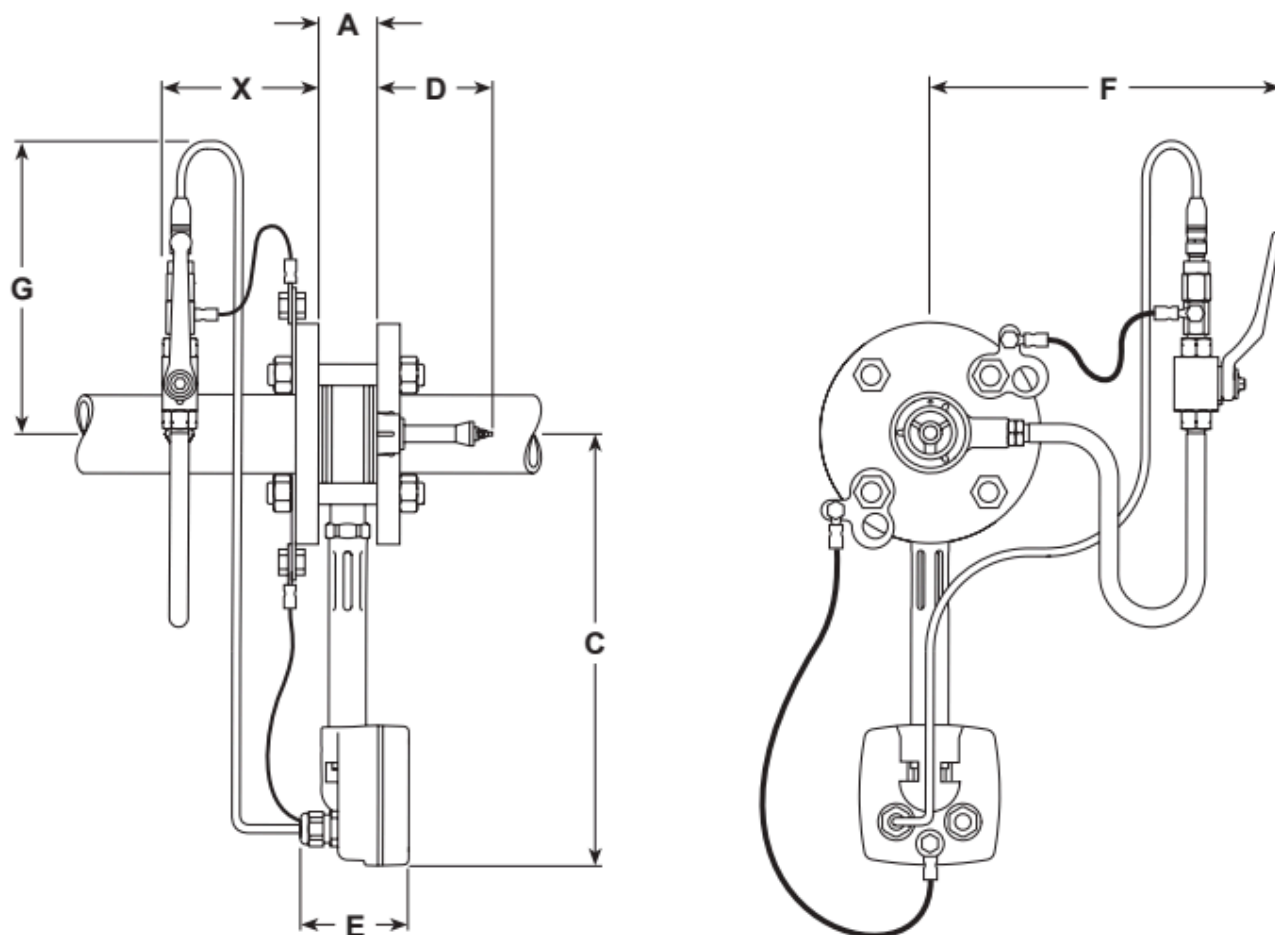
Technical data

IP rating	IP65 with correct cable glands
Power supply	Loop powered; with optional RS485: 24 VDC

Outputs	4–20 mA (not available with RS485 option); pulsed output (Vmax 28 Vdc, Rmin 10 kΩ)
Communication port	Modbus EIA 232C (RS 232C); with optional RS485: EIA 485 (RS 485C)

Dimensions/weights

Note: Dimension X is a recommended minimum distance between the pressure tapping and the flowmeter. It may be installed at any distance provided the cable allows. Standard cable length is 1 m.



Size	A	Flowmeter OD	C	D	E	F	G	X	TVA weight	Superheat kit	U syphon
DN50	35	103	322	125	65	250	160	300	2.67	0.3	0.5
DN80	45	138	334	115	65	270	160	300	4.38	0.3	0.5
DN100	60	162	344	155	65	280	160	300	7.28	0.3	0.5

Dimensions are approximate in mm and weights are approximate in kg.

Performance

The TVA flowmeter has inbuilt electronics which give a density compensated output. An LCD display is incorporated within the electronics head. The M750 display unit can provide a remote display function, using the 4–20 mA output.

System uncertainty, to 95% confidence (2 STD), in accordance with ISO 17025:

- $\pm 2\%$ of measured value from 10% to 100% of maximum rated flow.
- $\pm 0.2\%$ FSD from 2% to 10% of maximum rated flow.

- Turndown: up to 50:1.

As the TVA flowmeter is a self-contained unit, the uncertainty quoted is for the complete system. For technologies quoting pipeline-unit uncertainty, associated equipment such as DP cells must also be included to determine true system uncertainty.

Pressure drop

The pressure drop across the TVA is nominally 750 mbar (300 in water gauge) at maximum rated flow for DN50, and 500 mbar (200 in water gauge) for DN80 and DN100.

Flowmeter type	QE maximum (litres/min)	QE minimum (litres/min)	Maximum DP (in wg)	Maximum DP (mbar)
DN50	300	3	300	750
DN80	770	8	200	498
DN100	1,200	12	200	498

Sizing for saturated steam

Maximum flowrates in kg/h at different pressures (bar g), for horizontal orientation.

1. Maximum steam flowrates are calculated at maximum differential pressure.
2. The table is a guide only.
3. For superheated capacities, use the sizing software at www.spiraxsarco.com.

Size	Flow	1	3	5	7	10	12	15	20	25	30	32 bar g
DN50, QE 300	Max.	619	859	1,042	1,196	1,395	1,513	1,676	1,918	2,135	2,335	2,409
	Min.	12	17	21	24	28	30	33	38	43	47	60
DN80, QE 770	Max.	1,588	2,204	2,674	3,070	3,581	3,885	4,301	4,922	5,480	5,994	6,183
	Min.	32	44	53	61	72	78	86	98	110	120	128
DN100, QE 1,200	Max.	2,475	3,435	4,167	4,784	5,581	6,054	6,703	7,671	8,540	9,341	9,637
	Min.	49	69	83	96	112	121	134	153	171	187	192

Safety information, installation and maintenance

For full details see the Installation and Maintenance Instructions (IM-P192-02) supplied with the product. The following main points are given for guidance only:

1. Mount the TVA with at least 6 straight pipe diameters upstream and 3 downstream, with no valves, fittings or cross-sectional changes in those lengths. Allow 12 upstream diameters after an increase in nominal pipe diameter, two 90° bends in two planes, a pressure reducing valve or a partly open valve.
2. Ensure the internal upstream and downstream pipe diameters are smooth. Ideally use seamless pipe with no intrusive internal weld beads; slip-on flanges are recommended.
3. Install the TVA concentrically in the line to avoid flow measurement errors.
4. The TVA may be installed in any orientation up to 7 bar g for saturated conditions. For superheated conditions it may only be installed in horizontal pipework with the electronics below the pipeline.
5. Follow good steam engineering practice: provide correct line drainage and trapping, align and support pipework, use eccentric reducers for size changes, and do not insulate the TVA body or mating flanges.
6. Do not install the TVA outdoors where it may be exposed to driving rain or freezing.

How to order

Saturated service example: One Spirax Sarco DN100 TVA flowmeter for installation between EN 1092 PN40 flanges, for saturated steam at 10 bar g, maximum flow 5,581 kg/h.

Superheated service example: One Spirax Sarco DN100 TVA flowmeter, pressure sensing kit and U syphon, for installation between EN 1092 PN40 flanges, for superheated steam at 10 bar g.

Note: For details of the optional remote display, see the relevant Spirax Sarco M750 literature.

Spare parts and accessories

- Spare electronics front panel with standard RS 232C communications.
- Spare electronics front panel with RS485 communications converter.

Local regulations may restrict the use of this product to below the conditions quoted. In the interests of development and improvement of the product, Spirax Sarco reserves the right to change the specification without notice.

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